19.1 Capacitors

Question Paper

Course	CIE A Level Physics
Section	19. Capacitance
Topic	19.1 Capacitors
Difficulty	Medium

Time allowed: 40

Score: /26

Percentage: /100

Question 1a

Define the capacitance of a parallel plate capacitor.

[2 marks]

Question 1b

Two capacitors, of capacitances, C_A and C_B, are connected in parallel to a power supply of electromotive force (e.m.f.) E, as shown in Fig. 1.1.

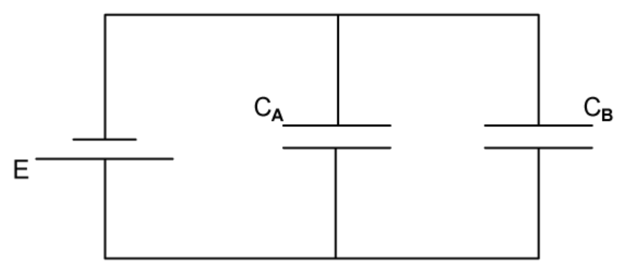


Fig. 1.1

Determine an expression for the combined capacitance C_C of the two capacitors.

[3 marks]

Question 1c

Two capacitors of capacitances 18 μF and 51 μF, and resistor of resistance 4.5 kΩ, are connected into the circuit of Fig. 1.2.

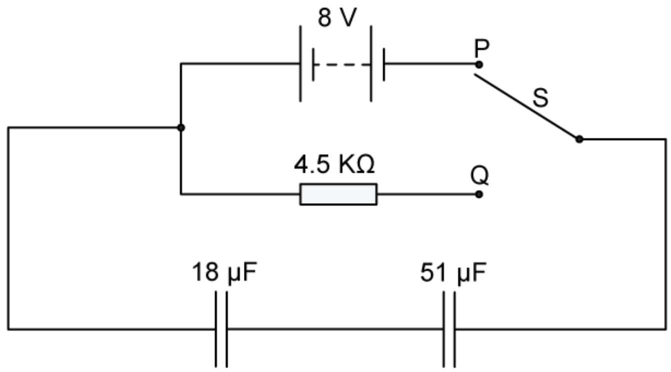


Fig. 1.2

The battery has an e.m.f of 8 V.

(i)
Calculate the combined capacitance of the two capacitors.

[1]

(ii)
The two-way switch S is initially at position P, so that the capacitors are fully charged.

Use the information in **(c) (i)** to calculate the total energy stored in the two capacitors.

total energy = ... J [2]

[3 marks]

Question 2a

Three capacitors, each of capacitance 27 μF, are connected as shown in Fig. 1.1.

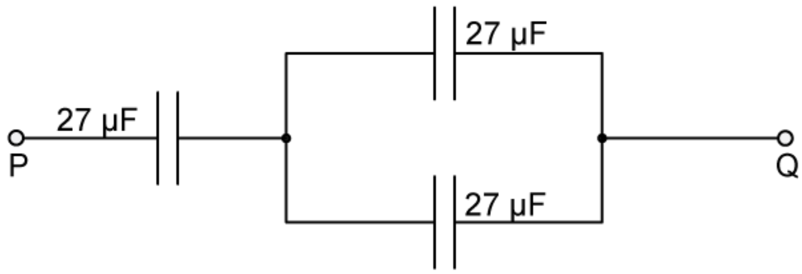


Fig. 1.1

Calculate the total capacitance between points P and Q.

capacitance = ... μF

[2 marks]

Question 2b

The maximum safe potential difference that can be applied across any one capacitor is 4 V.

Determine the maximum safe potential difference that can be applied between points P and Q.

potential difference = ... V

[2 marks]

Question 2c

A capacitor consists of an insulator separating two metal plates, as shown in Fig. 1.3.

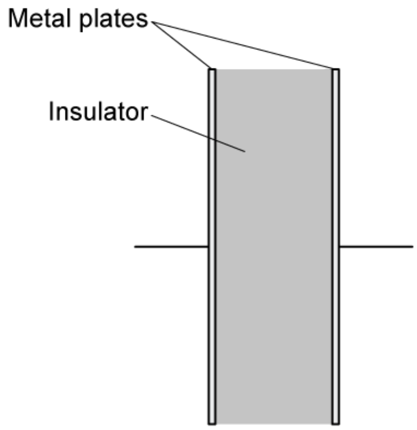


Fig. 1.3

The potential difference between the plates is *V*. The variation with *V* of the magnitude of the charge Q on one plate is shown in Fig. 1.4.

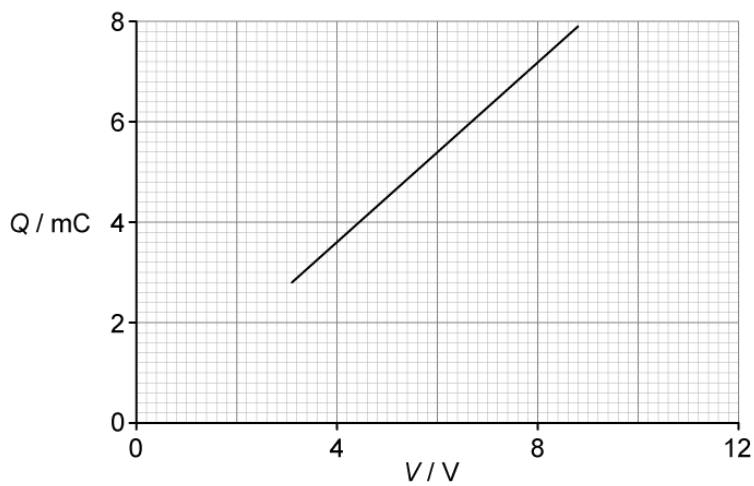


Fig 1.4

Explain why the capacitor stores energy but not charge.

[3 marks]

Question 2d

Use Fig. 1.4 to determine:

(i)

the capacitance of the capacitor

capacitance = ... μF [2]

(ii)

the loss in energy stored in the capacitor when the potential difference *V* is reduced from 8 V to 6 V.

energy = ... mJ [2]

[4 marks]

Question 3a

State two functions of capacitors connected in electrical circuits.

[2 marks]

Question 3b

Three capacitors are connected in parallel to a power supply as shown in Fig. 1.1.

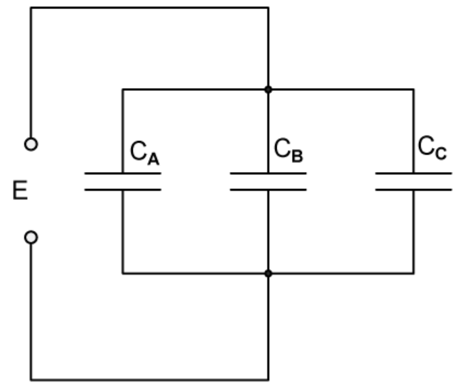


Fig. 1.1

The capacitors have a capacitance C_A, C_B and C_C. The power supply provides a potential difference *E*.

(i)

Explain why the charge on the positive plate of each capacitor is different.

[1]

(ii)

Use your answer in **(i)** to show that the combined capacitance C_T of the three capacitors is given by the expression.

$$C_T = C_A + C_B + C_C$$

[2]

[3 marks]

Question 3c

A student has available three capacitors, each of capacitance 24 μF.

Draw circuit diagrams, one in each case, to show how the student connects the three capacitors to provide a combined capacitance of:

(i)
36 μF

[1]

(ii)
16 μF

[1]

[2 marks]

Model Answers: Medium

1a

(a) The capacitance of a parallel plate capacitor is:

- Charge / potential difference; [1 mark]
- Where the charge is the charge on one plate, and the potential difference is across the two plates; [1 mark]

[Total: 2 marks]

1b

(b) Determine an expression for the combined capacitance C_C of the two capacitors:

Determine the p.d. across C_A and C_B:

- p.d. across both capacitors = E [1 mark]

Determine the total charge, Q_C in the circuit:

- $Q_C = Q_A + Q_B$ [1 mark]

Use the equation for p.d. and charge to determine an expression in terms of capacitance:

- $C = \frac{Q}{V}$ so $Q = VC$
- $EC_C = EC_A + EC_B$
- $C_C = C_A + C_B$ [1 mark]

[Total: 3 marks]

Remember that in a **parallel circuit**:

$$e.m.f. = p.d.1 + p.d.2....$$

$$Total\ charge = sum\ of\ charges\ across\ each\ branch$$

This is Kirchhoff's second law.

1c

(c)

List the known quantities:

- e.m.f., E = 8 V
- C_1 = 18 μF = 18 × 10^{-6} F
- C_2 = 51 μF = 51 × 10^{-6} F

(i) Calculate the combined capacitance of the two capacitors:

Obtain an expression for the total capacitance of the two capacitors in a series circuit:

- $\frac{1}{C_T} = \frac{1}{C_1} + \frac{1}{C_2}$
- $\frac{1}{18} + \frac{1}{51} = 0.075$
- $\frac{1}{0.075} = 13.3 \ \mu F = 13 \ \mu F \ (2 \ s.f.)$ [1 mark]

(ii) Calculate the total energy stored in the two capacitors:

- $energy = \frac{1}{2}CV^2$ [1 mark]
- $energy = \frac{1}{2} \times (13 \times 10^{-6}) \times 8^2$
- $energy = 4.16 \times 10^{-4}$ J [1 mark]

[Total: 3 marks]

2a

(a) Calculate the total capacitance between points P and Q:

List the known quantities:

- Capacitance of each capacitor, C = 27 μF

Calculate the capacitance of the parallel combination, C_P:

- $C_P = C + C$
- $C_P = 27 + 27$
- $C_P = 54\ \mu\text{F}$ [1 mark]

Calculate the capacitance of the series combination, C_S:

- $\frac{1}{C_S} = \frac{1}{C_P} + \frac{1}{C}$
- $\frac{1}{C_S} = \frac{1}{54} + \frac{1}{27} = \frac{1}{18}$
- $C_S = 18\ \mu\text{F}$ [1 mark]

[Total: 2 marks]

You do not need to convert the capacitance from μF into F because the final answer is required in μF.

2b

(b) Determine the maximum safe potential difference that can be applied between points P and Q:

List the known quantities:

- Max p.d. across each capacitor = 4 V
- Capacitance of each capacitor, $C = 27 \times 10^{-6}$ F (From part (a))
- Capacitance across the parallel combination, $C_P = 54 \times 10^{-6}$ F (From part (a))

Calculate the safe p.d. across the parallel combination:

- For components in series, the charge across each component is the same
  - Consider the single capacitor as component 1 and the parallel combination as component 2 in the circuit
  - Hence, $Q_1 = Q_2$
- The total charge from P to Q = charge across the single capacitor
  - $Q_1 = C_1 V_{max}$
  - $Q_1 = (27 \times 10^{-6}) \times 4$
  - $Q_1 = 1.08 \times 10^{-4}\ \text{C}$
- So, the p.d. across the parallel combination is:

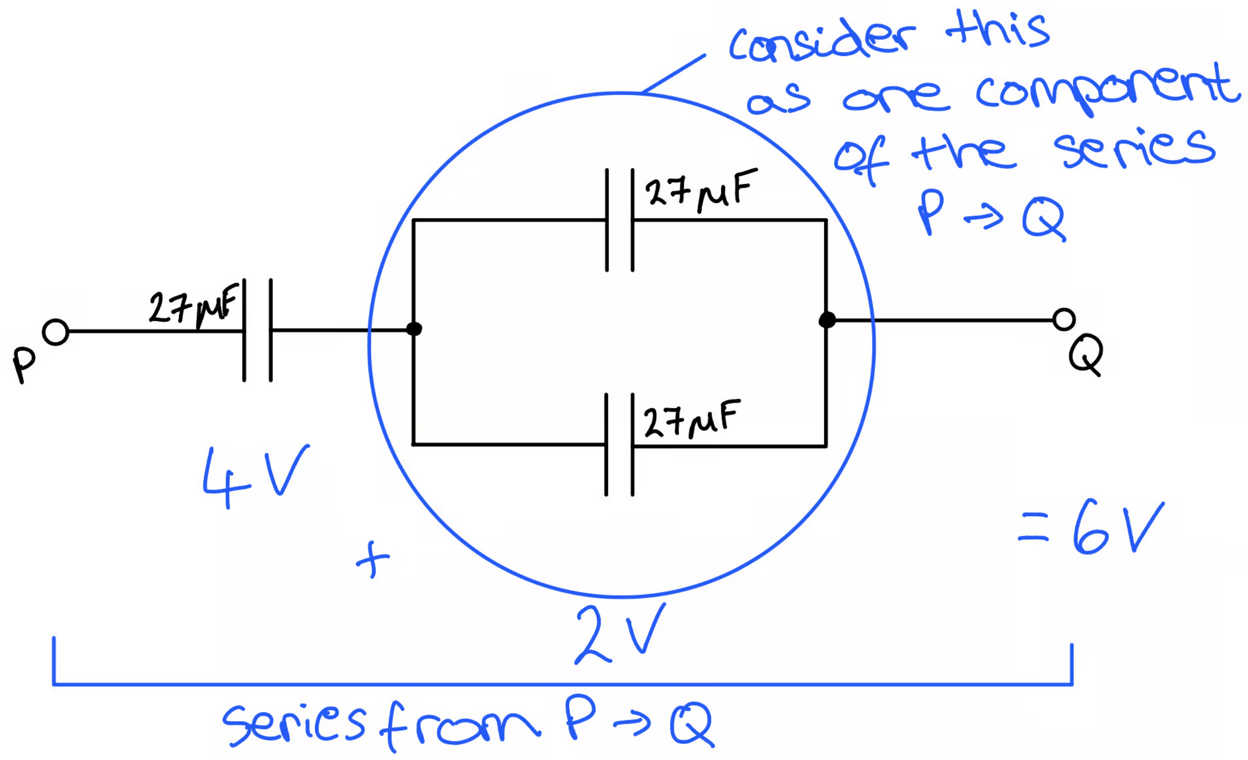
- $V_2 = \frac{Q_2}{C_P} = \frac{(1.08 \times 10^{-4})}{(54 \times 10^{-6})} = 2\ \mathrm{V}$ [1 mark]

Calculate the p.d. across the series combination:

- p.d. in series combination = p.d. in parallel + p.d. across single
- p.d. in series combination = 2 + 4 = 6 V [1 mark]

[Total: 2 marks]

If you're stuck on how the potential differences compare between the capacitors, this sketch might help:



2c

(c) The capacitor stores energy but not charge because:

- The charges on the plates of the capacitor are equal and opposite; [1 mark]
- So there is no resultant charge on the capacitor; [1 mark]
- But there is energy stored because there is a charge separation; [1 mark]

[Total: 3 marks]

2d

(d)

(c) The capacitor stores energy but not charge because:

- The charges on the plates of the capacitor are equal and opposite; [1 mark]
- So there is no resultant charge on the capacitor; [1 mark]
- But there is energy stored because there is a charge separation; [1 mark]

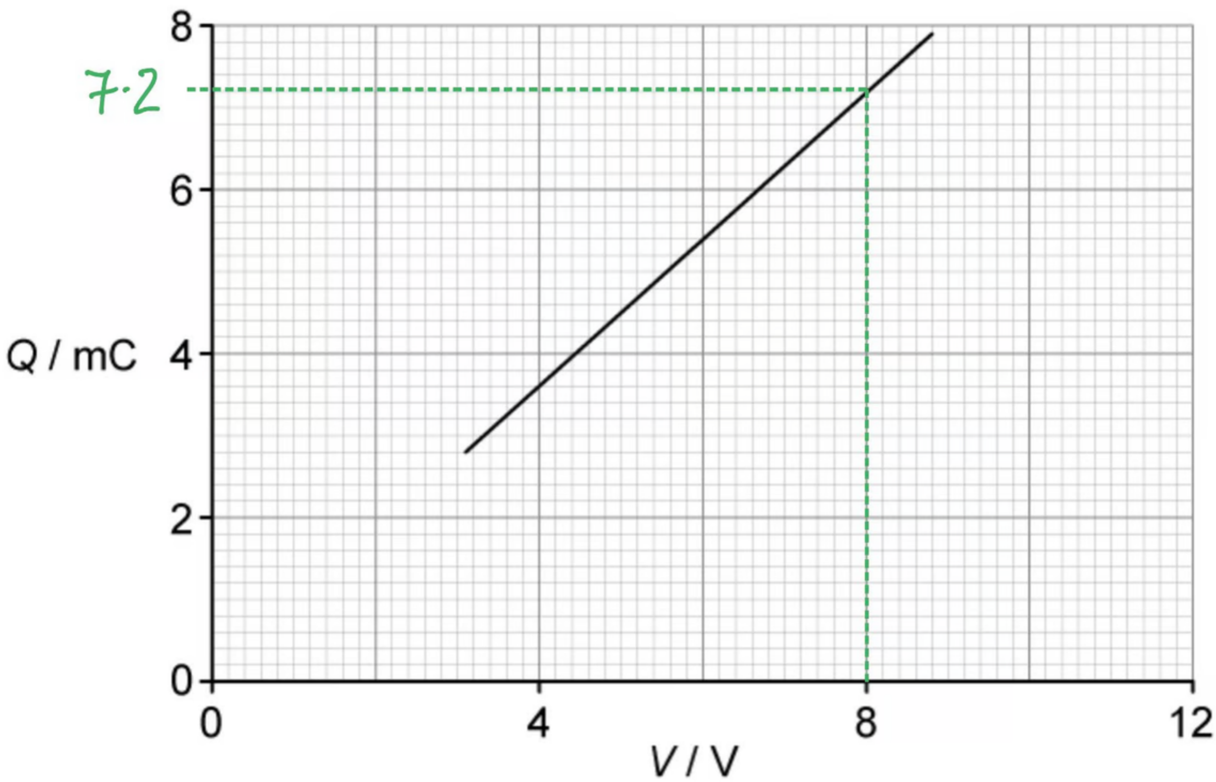
[Total: 3 marks]

2d

(d)

(i) Determine the capacitance on the capacitor:

Draw lines on the graph to obtain a set of coordinates from the graph:



- p.d., $V = 8$ V
- Charge, $Q = 7.2$ mC

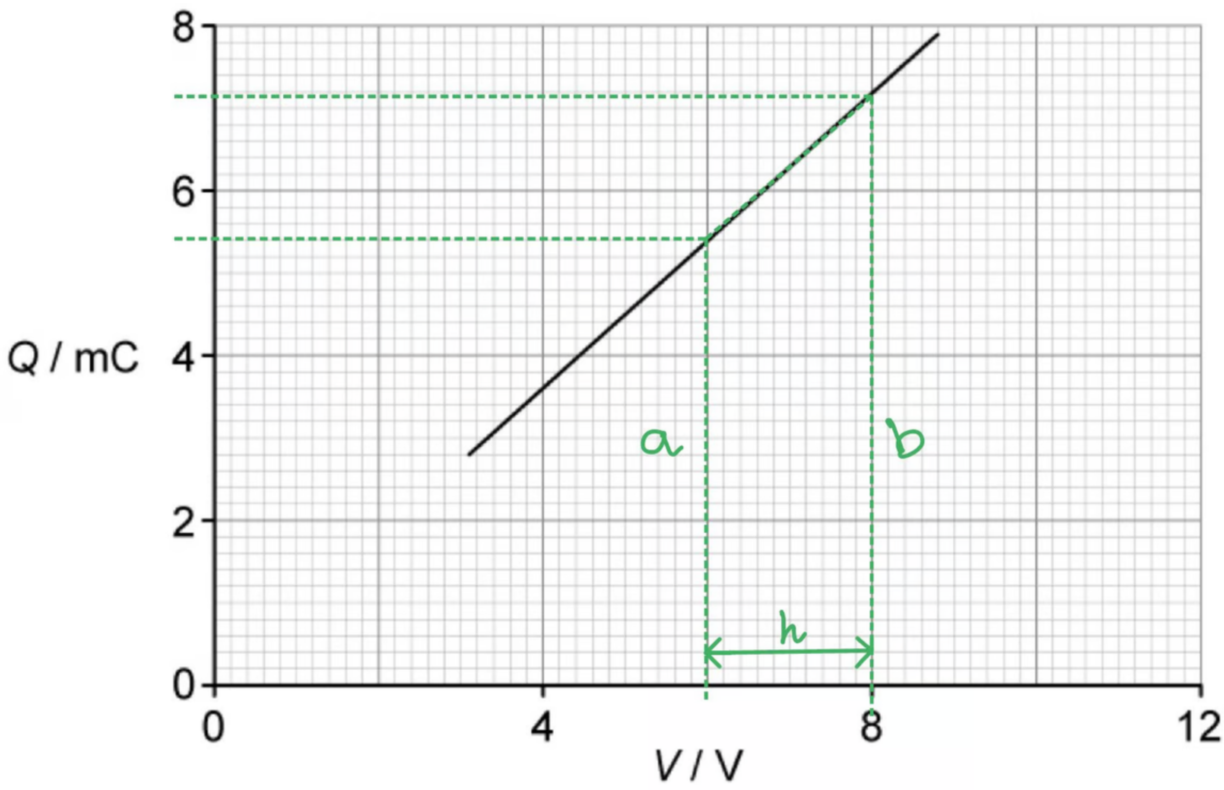
Calculate the capacitance:

- $C = \dfrac{Q}{V}$ [1 mark]

- $C = \dfrac{7.2 \times 10^{-3}}{8}$
- $C = 9 \times 10^{-4}\text{ F} = 900 \times 10^{-6}\ \mu\text{F}$ [1 mark]

(ii) Determine the loss in energy stored in the capacitor when the potential difference *V* is reduced from 8 V to 6 V:

Draw lines on the graph to identify the area of the trapezium underneath:



- $a = 5.4 \times 10^{-3}$ C
- $b = 7.2 \times 10^{-3}$ C
- $h = 2$ V

Calculate the area of the trapezium under the graph:

- $A = \frac{1}{2}(a + b)h = \frac{1}{2}((5.4 \times 10^{-3}) + (7.2 \times 10^{-3})) \times 6$ [1 mark]
- $A = 0.0378\text{ J} = 38\text{ mJ}$ [1 mark]

[Total: 4 marks]

You will be expected to remember the equation for the area of a trapezium (and other simple 2D shapes such as a right-angled triangle and rectangles) in your exam!

3a

(a) The functions of capacitors connected in electrical circuits are:

Any **two** from:

- To store energy; [1 mark]
- In smoothing circuits; [1 mark]
- To block direct current; [1 mark]
- In oscillators; [1 mark]

[Total: 2 marks]
To store **charge** is not an acceptable answer. Yes, a capacitor stores charge but it is not what it is used for in electrical circuits.

3b

(b)

(i) The charge on the positive plate of each capacitor is different because:

All **three** statements required for 1 mark:

- The potential difference across each capacitor is the same, E
- Each capacitor has a different capacitance
- $Q = CE$, so for each plate of the capacitor Q is different; [1 mark]

(ii) Show that the combined capacitance C_T of the three capacitors is given by the expression $C_T = C_A + C_B + C_C$:

Obtain an expression for the total charge across the capacitors in parallel:

- $Q_T = Q_A + Q_B + Q_C$ [1 mark]

Obtain an expression in terms of capacitance and p.d. for the total charge across the capacitors:

- $C_T E = C_A E + C_B E + C_C E$ [1 mark]
- Therefore, $C_T = C_A + C_B + C_C$

[Total: 3 marks]

The potential difference across each capacitor is the same **because** they are in parallel.

In part (ii), simplify the expression by **dividing** both sides **by *E*** to obtain the expression required.

3c

(c) Consider mathematically the combination of capacitors and their combined capacitance:

List the known quantities:

- Capacitance of each capacitor, $C = 24$ μF

Two capacitors in series with one in parallel:

- $C = \left(\frac{1}{\frac{1}{24} + \frac{1}{24}}\right) + 24 = 36 \text{ μF}$
- Hence, this is the arrangement needed for (i)

Two capacitors in parallel with one in series:

- $C = \frac{1}{\left(\frac{1}{24 + 24}\right) + \frac{1}{24}} = 16 \text{ μF}$
- Hence, this is the arrangement needed for (ii)

(i) A circuit diagram to show the 3 capacitors connected to obtain a capacitance of 36 μF is:

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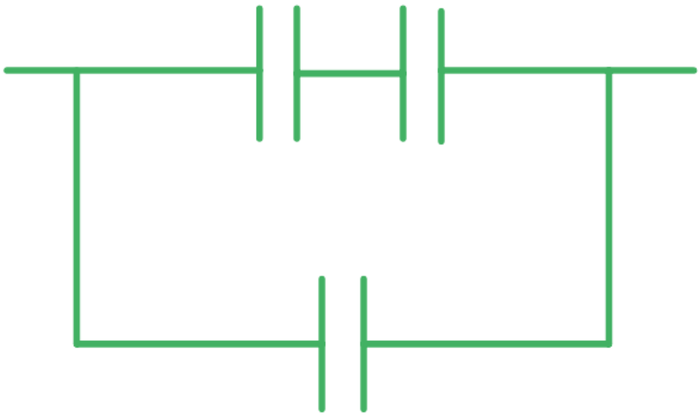
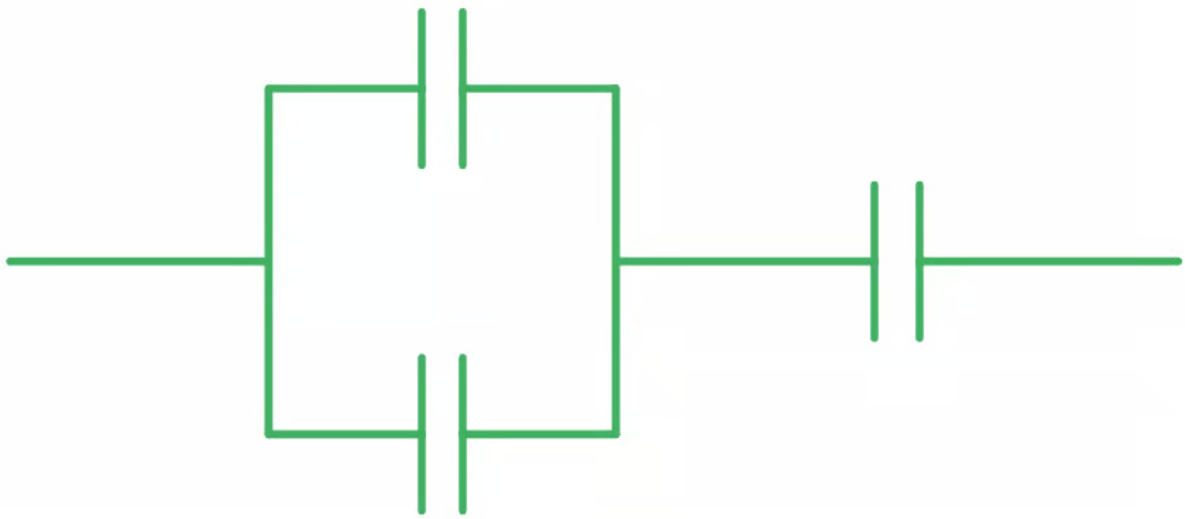


Two capacitors are drawn in series with one capacitor drawn in parallel to these; [1 mark]

(ii) A circuit diagram to show the 3 capacitors connected to obtain a capacitance of 16 μF is:



- Two capacitors are drawn in parallel with one capacitor in series to these; [1 mark]

[Total: 2 marks]

This question requires a bit of **trial and error** by considering the different combinations of capacitors in a logical order:

- All in parallel

- All in series
- Two in parallel and one in series - then mathematically you will notice this is an answer